Generative retrieval for next-item recommendation

Encode product content as Semantic IDs

Learn to generate the next item ID from a user history

One real-data CPU experiment; two simple baselines

พัสกร ยืนยง · 30 seconds

README.md; configs/poc01.yaml

Data and chronological holdout

Amazon Beauty 2014: 2,000 users and 7,154 items

17,264 events → 13,264 train events → 11,264 examples

[A,B,C] train; D validation; E test

Test history includes D; training targets exclude D and E

พัสกร ยืนยง · 60 seconds

src/data.py: make_subset; src/dataset.py: build_train_examples

From product content to unique Semantic IDs

Title, brand, category, price → Sentence-T5 vector [768]

Encoder 768 → 512 → 256 → 128 → 32

Three residual codebooks: 256 vectors each

Append collision suffix; freeze 7,154 unique IDs

พัสกร ยืนยง · 90 seconds

src/rqvae.py; src/semantic_ids.py; reports/poc01/id_diagnostics.json

Constructing model inputs and labels

Raw ID [236,227,111,0] → tokens [238,485,625,770]

Input: user bucket + latest 20 item IDs + EOS

Input length ≤82; target length =5; vocabulary =3,026

PAD=0; EOS=1; ignored label padding =−100

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src/tokenization.py: HistoryEncoder; src/dataset.py: collate

Training and selecting the recommender

Small T5 trained from scratch with teacher forcing

Labels [c0,c1,c2,c3,EOS]; input [PAD,c0,c1,c2,c3]

AdamW and mean token cross-entropy; 5,000 steps

Validation NDCG@10 selects step 4,500

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src/recommender.py; src/training.py; reports/poc01/experiment.json

Disclosed changes from the paper

Encoder/decoder: 4+4 → 2+2 layers; heads: 6×64 → 4×32

Feed-forward: 1,024 → 512; batch: 256 → 64

Beauty training: 200,000 → 5,000 steps; smaller subset

Quantizer: Adam + warm-up; retrieval: catalog prefix constraints

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README.md: settings table; docs/deviations.md; arxiv.org/html/2305.05065#S4

Generating and ranking catalog items

Beam width 20; generate four ID tokens and EOS

Trie permits catalog-valid continuations

Rank by summed token log-probabilities; return ten unique items

Generated tokens → raw SID → product lookup

ธิปก ปัญญาวุธ · 60 seconds

src/retrieval.py: CatalogTrie, Retriever; reports/poc01/example_trace.json

Two baselines and held-out results

Popularity: Recall@10 0.75%; NDCG@10 0.00279747

Item cosine: Recall@10 1.75%; NDCG@10 0.00655030

TIGER: Recall@10 2.35%; NDCG@10 0.01110530

Top-ten hits: 15 / 35 / 47 out of 2,000 users

ธิปก ปัญญาวุธ · 60 seconds

src/baselines.py; src/metrics.py; reports/baselines/results_test.json

What the results support and leave open

Pipeline verified; TIGER exceeds two simple baselines in this run

Absolute accuracy remains low: 1,953 top-ten misses

One seed; no random-ID ablation or significance test

36.4% of test targets lack training events; subgroup accuracy unknown

ธิปก ปัญญาวุธ · 60 seconds

reports/presentation_data_profile.json; reports/baselines/results_test.json

Actual contributions and shared preparation

6734446323 พัสกร ยืนยง: initialization and Developer A code

6734453723 ภาณุพงศ์ จันทร์ลาม: Developer B code

6634424423 ธิปก ปัญญาวุธ: code review and dataset cleanup

Every member prepares to explain all modules

All three · 60 seconds

docs/contributions.md

Backup: quantization and gradient routing

$c = \text{argmin} \, ||\text{residual} - \text{codebook}[j]||^2$

Residual subtracts the selected vector at each level

Forward: z + (quantized – z).detach() equals quantized

Backward: reconstruction reaches encoder; auxiliary losses train codebooks

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src/rqvae.py: quantize, forward; tests/test_quantizer.py

Backup: attention and tensor shapes

Input IDs and attention mask: [batch,L], L≤82

Labels: [batch,5]; logits: [batch,5,3026]

Encoder self-attention uses observed history

Decoder: causal self-attention plus encoder cross-attention

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src/recommender.py: build_model; src/dataset.py: collate

Backup: baseline worked example

Training users: U1={a,b}; U2={a,c}

n(a)=2; n(b)=1; shared(a,b)=1

sim(a,b)=1/√2; sim(b,c)=0

Scores sum similarities; zero-score ties use training popularity

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src/baselines.py: ItemCosine; tests/test_item_cosine.py

Backup: ranking scores and evaluation

Beam score = Σ log P(token | history, prefix)

Constraint masking does not prove globally optimal top K

Target at rank 3: Recall@5=1; NDCG@5=0.5

Empty predictions remain in averages; duplicates removed

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tests/test_model_retrieval.py; src/metrics.py

Backup: frozen checkpoints and recovery

best/ stores validation-selected inference weights

latest.pt stores model, optimizer, RNG, epoch, and batch cursor

Mapping and input hashes guard against incompatible restores

59 tests pass; original experiment recorded 57

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src/training_state.py; tests/test_training_resume.py